

Alpilla 600 MW Combined Cycle Power Plant

Painting and Coating Specification

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Originator	Istanbul EPC Mühendislik ve Taahhüt A.Ş.
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Rev	Purpose	Date	Description	Prepared	Checked	Approved
A	IFR	2026-12-14	First issue for Owner approval	EPCE	EPC mechanical lead	EPC engineering manager

1. Purpose and scope

1.1 This specification defines the surface preparation, coating systems, application, colours, inspection and hold points for painting of steel structures, equipment, piping and tanks of the power plant, for the corrosivity category of the site (C5, durability H).

1.2 The specification applies to coating applied in the shop and on site to carbon and low-alloy steel, galvanised steel, and to stainless steel where a coating is required. It does not apply to the internal linings of seawater equipment (rubber lining) defined in the package specifications, nor to the finish of GT, ST and generator packages of IEC, which shall be equivalent and approved by IEPC.

2. Governing requirements and site data

ER-02.08: Outdoor equipment and structures shall be designed for corrosivity category C5 (ISO 12944-2) with a coating system of high durability (H, more than 15 years to first major maintenance).

ER-13.03: Structural steel per EN 1993 with seismic detailing, hot-dip galvanised outdoors (EN ISO 1461).

Parameter	Value	Condition	Status	Basis
100-year coastal still water level	0.9 m	national datum; +1.40 m with 0.5 m sea-level rise; floor levels $\geq$ +1.90 m (ER-02.09); shoreline run-up +4.8 m	confirmed	SRC-OWNR-0002, ER-02.09
Atmospheric corrosivity category	C5 (ISO 12944-2), durability H -		confirmed	ER-02.08
Basic wind speed	30 m/s	10 min mean, 10 m, 50-year return	confirmed	ER-02.05
Design rainfall intensity	110 mm/h	10 min, 25-year return	confirmed	ER-02.05
Distance to heavy-lift port	18 km by sea	400 t quay crane	confirmed	SRC-OWNR-0005
Ground snow load	0.75 kN/m ²		confirmed	ER-02.05
Groundwater level	13.5 m a.s.l.	brackish, chloride 3,000-8,000 mg/l	confirmed	SRC-OWNR-0008
Heavy transport road limit	120 t gross	two bridges on the state road	confirmed	SRC-OWNR-0005, ER-02.14
Height datum	TUDKA-99 orthometric heights (m a.s.l.)		confirmed	SRC-EPCE-0011
Plant grid origin in UTM zone 35N (E 0 / N 0)	E 541,250.000 m / N 4,532,480.000 m	TUREF (ITRF96), south-west plot corner	confirmed	SRC-EPCE-0011
Platform level acceptance tolerance	0.15 m	Owner-levelled platform +15.00 m	confirmed	DEC-OWNR-0002
Plot dimensions	400 x 300 m	12.0 ha; usable about 10.6 ha	confirmed	SRC-OWNR-0005, ER-02.01
Rotation plant grid to UTM grid	-0.3281 deg	plant north = true north; grid convergence at site	confirmed	SRC-EPCE-0011
Scale factor plant grid to UTM	0.999612 -	combined UTM point scale and height factor	confirmed	SRC-EPCE-0011
Seismic design PGA (DD-2, 475-year)	0.45 g	TBDY 2018, site-specific hazard study	confirmed	ER-02.06
Seismic site class	ZD (Vs30 = 240 m/s) -	TBDY 2018	confirmed	SRC-OWNR-0008
Site grade level	15 m a.s.l.		confirmed	ER-02.01

2.1 Chloride deposition and salt aerosol values of DP-OWNR-0005, DP-OWNR-0006 and DP-OWNR-0007 are the basis of the systems chosen; systems shall be qualified per ISO 12944-6 for C5 (laboratory tests) or shall have documented field experience of at least 10 years in marine C5.

3. Codes and definitions

3.1 Coating shall follow the series of ISO 12944 (parts 1 to 9, corrosivity per ISO 12944-2), ISO 8501 (rust grades and preparation grades), ISO 8502 (tests for surface cleanliness), ISO 8503 (surface profile), ISO 19840 (dry film thickness measurement) and ISO 2808. Galvanising per EN ISO 1461 and duplex systems (galvanising and coating) per ISO 12944-5.

3.2 Durability is class H (more than 15 years to first major maintenance) for all coating systems of outdoor items; interior surfaces of buildings are class C3 (medium) unless the room is classified higher.

3.3 The coating manufacturer shall be approved by IEPC; all coats of one system shall be from the same manufacturer. Coatings shall be free of lead and chromates, and shall satisfy the Turkish VOC regulations.

4. Surface preparation

4.1 Carbon steel surfaces to be coated shall be blast cleaned to Sa 2.5 (ISO 8501-1) with angular abrasive; surface profile 50 to 85 micrometres (medium, ISO 8503, comparator G or S) for systems with zinc-rich primer, 40 to 70 micrometres for other systems. Weld spatter, sharp edges (radiused to at least 2 mm) and laminations shall be removed prior to blasting (ISO 8501-3 grade P3).

4.2 The steel surface temperature during blasting and coating shall be at least 3 K above the dew point, and relative humidity shall not exceed 85 %; abrasive shall be free from chlorides (conductivity of the water extract not above 250 microsiemens/cm) and soluble salts on the surface shall not exceed 20 mg/m² of chloride (ISO 8502-6/9).

4.3 Blast cleaning shall be applied within 4 hours of cleaning and before any visible rust (flash rust); surfaces with rust re-formed shall be re-blasted. Power-tool cleaning to St 3 is accepted only for touch-up and small areas.

4.4 Galvanised surfaces shall be cleaned of oil and salts, sweep blasted (roughening, with no removal of the zinc layer) or treated with a mordant primer when they are to be coated; the minimum zinc thickness shall be per EN ISO 1461.

4.5 Stainless steel surfaces shall not be blasted with carbon steel grit; the surfaces shall be degreased and coated only with chloride-free coatings; contamination by iron shall be removed by pickling.

5. Coating systems

System	Specification	Remarks
Carbon steel, outdoor, service temperature up to 120 degC, durability H (C5)	Zinc-rich epoxy primer 60 to 80 micrometres + epoxy MIO / high-build epoxy 120 to 160 micrometres + polyurethane top coat 50 to 60 micrometres; total DFT 280 micrometres minimum	ISO 12944-5 system for C5 high durability; total nominal DFT 320 micrometres is the design target [REF-CONS-0092]
Carbon steel in the splash zone, seawater intake structures, pump house metalwork (CX / Im2)	Zinc-rich epoxy primer 60 micrometres + 2 x high-build epoxy (glass flake) 300 micrometres each; total DFT 700 micrometres minimum	Combined with cathodic protection where immersed [REF-CONS-0092, ALP-EPC-00000-EL-CAL-0005]
Carbon steel, hot surfaces from 120 degC to 400 degC, uninsulated	Inorganic zinc silicate or silicone aluminium; total DFT 50 to 100 micrometres	Temperature range per the coating manufacturer's data sheet
Carbon steel, hot surfaces above 400 degC, and stainless steel under insulation	Thermally sprayed aluminium (TSA) 200 micrometres minimum with sealer, per ISO 2063 (or a silicone aluminium alternative above 400 degC)	TSA is the default under insulation on carbon steel and stainless steel at 60 to 595 degC, to avoid corrosion and chloride stress corrosion cracking under insulation [ALP-EPC-00000-ME-SPC-0003]
Carbon steel, cold surfaces and insulated below 120 degC	Epoxy phenolic or epoxy 2 x 100 micrometres (total DFT 200 micrometres) under insulation for temperatures from -10 to 120 degC	Cyclic services get TSA; refer to the insulation specification [ALP-EPC-00000-ME-SPC-0003]
Galvanised steel, outdoor, structural, gratings	Hot-dip galvanising per EN ISO 1461 (minimum 85 micrometres) plus epoxy / polyurethane duplex system 100 to 160 micrometres in C5	Duplex system per ISO 12944-5; ER-13.03 requires galvanised outdoor structural steel [REF-CONS-0091, ER-13.03]
Storage tanks, external (LDO, fire water)	Zinc-rich epoxy primer 60 micrometres + epoxy 160 micrometres + polyurethane 60 micrometres; roof and shell as above; tank bottom underside epoxy 300 micrometres total	Tanks per API 650; internal coating per the tank specification [REF-CONS-0066]
Tanks and vessels, internal (LDO, demin water)	Epoxy or phenolic lining 300 to 500 micrometres, holiday tested	Internal coating or linings for LDO and potable water tanks approved for the service
Stainless steel (where coating required)	Epoxy 2 x 80 micrometres after cleaning, no blast with iron grit	Coating on stainless steel only when required by temperature and chloride conditions

System	Specification	Remarks
Buried carbon steel piping	3-layer polyethylene or polypropylene (external) or epoxy 400 micrometres plus wrapping; cathodic protection	Only where GRP piping is not used [ALP-EPC-00000-EL-CAL-0005]
Electrical and instrument enclosures, cable trays (steel)	Polyester powder coat 80 micrometres over zinc primer, or hot-dip galvanising; C5 tested	Equipment register states C5 marine coating for switchgear and cubicles

6. Application and dry film thickness

6.1 Coating shall be applied by airless spray; brush or roller application is restricted to stripe coats and touch-up. A stripe coat shall be applied on all edges, welds, bolts, nuts and other difficult areas before the full coat of each of the first two coats.

6.2 The dry film thickness (DFT) shall be measured to ISO 19840; the '80/20' rule applies: 80 % of all readings shall be equal to or above the nominal DFT and no single reading shall be lower than 80 % of the nominal DFT; the maximum DFT shall not exceed 3 times the nominal DFT.

6.3 Overcoating intervals and pot life shall follow the paint manufacturer's data sheet; where the interval is exceeded, the surface shall be roughened and cleaned. Coating shall not be applied when the temperature of the steel is below 5 degC (epoxy) or below the manufacturer's minimum, or when rain, fog or salt spray wets the surface.

6.4 Shop coating shall cover all coats except the final top coat (or the final coat of the system where the item is shipped in the finish colour); field welds and damaged areas shall be repaired to the same system, including the surface preparation grade St 3 or Sa 2.5 in the touch-up area.

6.5 Areas that shall not be coated: machined surfaces, nameplates, stainless steel instrument tubing, flange faces, gasket surfaces, bolt threads (only greased), and rotating shafts. They shall be masked and protected with removable preservatives during coating.

7. Colours

7.1 Colours shall follow RAL codes per service: process piping colour bands by service with the base colour of the structure; structural steel RAL 7035 (light grey); equipment RAL 7035 with service-coloured pipes below.

Service	Colour (RAL)	Remarks
Fire water piping and equipment	RAL 3000 (fire red)	Safety colours per ISO 3864
Seawater / cooling water piping	RAL 5015 (sky blue) with green band for seawater	
Demineralised, potable and process water	RAL 5005 (signal blue); potable RAL 5012 / green band	
Steam and hot water piping (insulated: cladding colour; uninsulated: coating)	RAL 3020 (traffic red) with service bands	
Fuel gas	RAL 1021 (yellow) with band for gas	
LDO and other oils	RAL 8003 (brown)	
Compressed air and nitrogen	RAL 5012 (air, light blue) / RAL 6018 (nitrogen, green)	
Chemicals (acid / caustic / ammonia)	RAL 2004 (pure orange) with violet band for ammonia	Hazard signs per Turkish regulations
Drains	RAL 9005 (black)	

7.2 Hazard marking of equipment (yellow and black bands, safety signs) shall follow ISO 3864 and ISO 7010; colour codes of piping shall be shown on the piping and instrumentation diagrams and in the piping design criteria.

8. Inspection, hold points and documentation

8.1 The coating contractor shall issue a coating procedure specification and a quality plan per product and system, approved by IEPC before start. A procedure qualification on a test panel (all coats, DFT, adhesion) shall be carried out for each system.

Hold / witness point	Inspection	Remarks
H1 - before blasting	Surface condition, contamination (weld spatter, oil), edge treatment inspected	Hold point [ER-15.02]
H2 - after blasting	Sa 2.5 degree, profile and soluble salt content measured; weather conditions recorded	Hold point before priming [ER-15.02]
W3 - after primer and after each coat	DFT measured, visual defects, stripe coat check	Witness point [ER-15.02]

Hold / witness point	Inspection	Remarks
H4 - final coat	Total DFT, holiday detection for immersed and lined surfaces, adhesion pull-off test (ISO 4624) with minimum 5 MPa (3 MPa for TSA per ISO 2063)	Hold point before shipping or release [ER-15.02, REF-CONS-0092]

8.2 Defects shall be repaired according to the paint manufacturer's instructions and re-inspected. Coating records (weather, surface preparation, batch numbers, DFT results, repairs) shall form part of the equipment dossier delivered to the Owner.

8.3 A warranty of 2 years for the coating system shall apply from the taking-over, with a maintenance-free expectation of more than 15 years to the first major maintenance (durability H); the coating manufacturer shall confirm the system's suitability for C5.

References

Codes and standards cited in this document (Codes, Standards and Regulations Register ALP-EPC-00000-GE-LST-0001; latest edition at contract date unless stated).

Code	Title
API 650	Welded Tanks for Oil Storage
EN ISO 1461	Hot dip galvanized coatings on fabricated iron and steel articles
ISO 12944-2	Paints and varnishes - Corrosion protection of steel structures by protective paint systems - Part 2: Classification of environments